

FLIPKART

The version of GCC being used is 5.5.0.

A state consists of N cities numbered from 0 to $N-1$. All the roads in the state are bidirectional. Each city is connected to another city by one direct road only. A magician travels to these cities to perform shows. He knows a magic spell that can completely eliminate the distance between any two directly connected cities. But he must be very careful because this magic spell can be performed only K number of times.

Write an algorithm to find the length of the shortest route between two given cities after performing the magic spell K number of times. The output is -1 if no path exists.

Input

The first line of the input consists of an integer - $numCities$, representing the number of cities (N).

The second line consists of an integer

- $cityA$, representing the city A.

The third line consists of an integer

- $cityB$, representing the city B.

The fourth line consists of an integer

Test Cases & Output

```
1 // Header
2 #include <bits/stdc++.h>
3 #include <iostream>
4 #include <vector>
5 using namespace std;
6
7
8 // Function to find the shortest path
9 * numCities
10 cityA
11 cityB
12 numMagic
13 cityC
14 */
15 int main()
16 {
17
18
19
20
21
22 }
23
24 int
25 {
26
27
28
29
30
31
32
```

Input

The first line of the input consists of an integer - $numCities$, representing the number of cities (N).

The second line consists of an integer - $cityA$, representing the city A.

The third line consists of an integer

- $cityB$, representing the city B.

The fourth line consists of an integer

- $numMagic$, representing the number of times the magician can perform the magic spell (K).

The fifth line consists of two space-separated integers - $numRoads$ and

$roadProp$, representing the number of roads

(M) and number of given properties of each road ($roadProp$ (R) is always equal to 3),

respectively.

The next M lines consist of R space-separated

integers - $cityU$, $cityV$ and $lenW$, where

$cityU$ and $cityV$ represent the cities

and $lenW$ represents the length of the

bidirectional road between the cities.

Output

Print an integer representing the length of the

Test Cases & Output

```
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3 #include <iostream>
4 #include <vector>
5 using namespace std;
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8 // Function to find the shortest path
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18
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21
22 }
23
24 int
25 {
26
27
28
29
30
31
32
```


Output

Print an integer representing the length of the shortest route between the two given cities after performing the magic spell K number of times. Print -1 if no path exists.

Constraints

$$1 \leq \text{numCities} \leq 1000$$

$$0 \leq \text{numMagic} \leq \text{numCities}$$

$$0 \leq \text{cityA}, \text{cityB} < \text{numCities}$$

$$0 \leq \text{numRoads} \leq 10^4$$

$$1 \leq \text{lenW} \leq 1000$$

Example

Input:

5
0
3
1
5 3
0 1 1
0 4 1
1 2 2
2 3 4
4 3 7

Test Cases

1 //
2 #
3 #
4 #
5 u
6
7
8 //
9
10 c
11 c
12 r
13 c
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
22

5
0
3
1
5 3
0 1 1
0 4 1
1 2 2
2 3 4
4 3 7

Output:

1

Explanation:

There are two possible routes between 0 and 3:
0->1->2->3 and 0->4->3

After reducing the distance of edge 4->3 to zero, the second route becomes 0->(4,3) and thus the minimum distance is 1.

```
1 // Header Files
2 #include<iostream>
3 #include<string>
4 #include<vector>
5 using namespace std;
6
7
8 //
9 * numCities is
10 cityA is the fir
11 cityB is the sec
12 numMagic is the
13 cityConn is a gr
14 */
15 int shortRoute
16 {
17     int answer
18     // Write yo
19
20
21     return ans
22 }
23
24 int main()
25 {
26     //input fo
27     int numCit
28     cin >> num
29
30     //input f
31     int cityA
32 }
```

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QUESTION

Work in case of syntax/runtime error.
The version of GCC being used is 5.5.0.

Jacklin, a math teacher at MathWorld, is preparing to administer a test to her class. The students will evaluate an equation that consists of "+", "-", and "*" operators. She will award a grade of '5' for a correct answer and a grade of '2' for a partially-correct answer featuring a mistake in the application of the BODMAS rule. If a student's answer is wholly incorrect, the student will receive a grade of '0.'

Write an algorithm to help Jacklin find the total class score for the given question, assuming every student has answered the question.

Input

The first line of the input consists of a string S , representing the equation to be evaluated.
The second line consists of an integer N , representing the total number of students.
The last line consists of N space-separated integers - $arr_1, arr_2, \dots, arr_N$ representing the answers given by the students.

Test Cases & Output

```
1 // Sample code to read input
2
3 /*
4  *include <iostream>
5  *using namespace std;
6  *int main()
7  *{
8  *    char name[20];
9  *    cin >> name;
10 *    cout << "Hello " << name << endl;
11 *    return 0;
12 *}
13 */
14
15 // Warning: Print the output
16
17 #include <iostream>
18 using namespace std;
19
20 int main()
21 {
22     // Write your code here
23     return 0;
24 }
```

Input

The first line of the input consists of a string S , representing the equation to be evaluated.
The second line consists of an integer N , representing the total number of students.
The last line consists of N space-separated integers - $arr_1, arr_2, \dots, arr_N$ representing the answers given by the students.

Output

Print an integer representing the total class score for the given equation.

Constraints

$0 \leq N \leq 1000$

$1 \leq \text{len} \leq 30$; where len is the length of S

$-10^9 \leq arr_1, arr_2, \dots, arr_N \leq 10^9$

Note

The string is the combination of number and operators +, - and *.

A string always begins and ends with a number.

No two operators occur consecutively in the

Test Cases & Output

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17 #include <iostream>
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```


QUESTION

Note

The string is the combination of number and operators +, - and *.

A string always begins and ends with a number.

No two operators occur consecutively in the equation.

Example

Input:

5*4-3*2

4

-10 10 14 90

Output :

9

Explanation :

Step1: According to BODMAS, correct answer = 14

Partially-correct answer = -10, 10, 34
Step2: Grades awarded to the students by the teacher = 2, 2, 5, 0

Step3: So, total grades for the class is 2+2+5
So, the output is 9.

Test Cases & Output

Editor

```
1 // Sample code to read input
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5  * using namespace std;
6  * int main()
7  * {
8  *     char name[20];
9  *     cin >> name;
10    *     cout << "Hello " << name;
11    *     return 0;
12    * }
13    */
14
15 // Warning: Printing
16
17 #include <iostream>
18
19 using namespace std;
20
21 int main()
22 {
23     // Write your code here
24     return 0;
25 }
```



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